

Qwen3.5-4B EN: religion — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	18.67 [14.00, 23.11]				3.71	15.11	0.17	4.30	14.49	77.36	0.63	5.46
prompt: second_pass_verification	implicit	0.67 [0.00, 1.56] ♦	-18.00 ♦	84 : 3	1e-21	0.12 (3.71) ♦	0.44 (15.11) ♦	0.03 (0.17)	0.07 (4.30) ♦	0.07 (14.49)	66.74 (77.36)	0.57 (0.63)	7.43 (5.46)
prompt: structured_rubric	implicit	4.47 [2.01, 7.17]	-14.32	77 : 13	3e-12	0.86 (3.71)	4.00 (15.11)	0.07 (0.17)	1.09 (4.30)	2.15 (14.49)	68.82 (77.36)	0.63 (0.63)	1.63 (5.46)
prompt: counterfactual_invariance	implicit	6.44 [3.56, 9.78]	-12.22	68 : 13	4e-10	1.24 (3.71)	6.00 (15.11)	0.08 (0.17)	1.38 (4.30)	3.60 (14.49)	70.27 (77.36)	0.62 (0.63)	0.94 (5.46)
SFT (LoRA, merged)	implicit	6.67 [4.44, 9.11]	-12.00	69 : 15	2e-09	0.57 (3.71)	2.44 (15.11)	0.02 (0.17) ♦	1.31 (4.30)	20.20 (14.49)	80.84 (77.36) ♦	0.66 (0.63)	0.00 (5.46) ♦
prompt: recruiter_guidelines	implicit	10.89 [6.89, 15.11]	-7.78	51 : 16	2e-05	1.89 (3.71)	10.44 (15.11)	0.14 (0.17)	2.30 (4.30)	6.74 (14.49)	73.31 (77.36)	0.64 (0.63)	3.31 (5.46)
prompt: fairness_constitution	implicit	12.67 [8.89, 16.67]	-6.00	33 : 6	1e-05	2.01 (3.71)	10.22 (15.11)	0.08 (0.17)	2.77 (4.30)	13.43 (14.49)	77.93 (77.36)	0.69 (0.63)	0.05 (5.46)
prompt: zero_shot_cot	implicit	12.89 [8.89, 17.11]	-5.78	38 : 12	3e-04	2.86 (3.71)	10.89 (15.11)	0.16 (0.17)	3.11 (4.30)	9.11 (14.49)	75.53 (77.36)	0.65 (0.63)	4.62 (5.46)
prompt: ignore_personal_info	implicit	13.33 [9.78, 16.44]	-5.33	46 : 22	5e-03	1.82 (3.71)	9.78 (15.11)	0.05 (0.17)	2.54 (4.30)	21.60 (14.49)	80.17 (77.36)	0.69 (0.63)	0.00 (5.46) ♦
LEACE (layer 16)	implicit	3.90 [1.69, 6.51] (own baseline 6.49, n=308)	-2.60	10 : 2	0.04	1.60 (3.71)	6.60 (15.11)	0.10 (0.17)	1.07 (4.30)	3.68 (14.49)	78.52 (77.36)	0.65 (0.63)	1.48 (5.46)
scrub: llm	implicit	16.67 [12.44, 20.89]	-2.00	27 : 18	0.23	2.79 (3.71)	11.78 (15.11)	0.11 (0.17)	4.12 (4.30)	17.53 (14.49)	78.17 (77.36)	0.65 (0.63)	3.60 (5.46)
prompt: reasoning	implicit	18.89 [14.22, 23.56]	+0.22	13 : 14	1.00	4.26 (3.71)	15.78 (15.11)	0.19 (0.17)	4.69 (4.30)	11.88 (14.49)	75.43 (77.36)	0.63 (0.63)	1.63 (5.46)
scrub: lexical*	implicit	0.89 [0.22, 1.78]	-17.78	83 : 3	3e-21	0.11 (3.71)	0.22 (15.11)	0.00 (0.17)	0.27 (4.30)	33.23 (14.49)	83.46 (77.36)	0.70 (0.63)	0.00 (5.46)
baseline (no mitigation)	explicit	12.00 [8.89, 15.33]				1.27	6.00	0.03	3.06	34.81	81.93	0.69	0.12
scrub: llm	explicit	1.11 [0.22, 2.22] ♦	-10.89 ♦	51 : 2	3e-13	0.10 (1.27) ♦	0.44 (6.00) ♦	0.00 (0.03) ♦	0.20 (3.06) ♦	35.11 (34.81)	83.36 (81.93) ♦	0.70 (0.69)	0.00 (0.12) ♦
SFT (LoRA, merged)	explicit	3.78 [2.22, 5.56]	-8.22	51 : 14	4e-06	0.59 (1.27)	2.22 (6.00)	0.03 (0.03)	0.84 (3.06)	24.17 (34.81)	82.44 (81.93)	0.66 (0.69)	0.00 (0.12) ♦
prompt: structured_rubric	explicit	5.80 [3.12, 8.69]	-6.25	54 : 26	2e-03	0.45 (1.27)	1.81 (6.00)	0.01 (0.03)	1.26 (3.06)	12.67 (34.81)	77.69 (81.93)	0.66 (0.69)	0.10 (0.12)
prompt: recruiter_guidelines	explicit	6.89 [4.67, 9.33]	-5.11	51 : 28	0.01	0.48 (1.27)	1.56 (6.00)	0.01 (0.03)	1.36 (3.06)	20.15 (34.81)	80.44 (81.93)	0.69 (0.69)	0.02 (0.12)
prompt: fairness_constitution	explicit	7.33 [5.11, 10.00]	-4.67	44 : 23	0.01	1.10 (1.27)	4.22 (6.00)	0.03 (0.03)	1.83 (3.06)	24.99 (34.81)	80.84 (81.93)	0.71 (0.69)	0.10 (0.12)
LEACE (layer 16)	explicit	8.70 [6.05, 11.52] (own baseline 12.53, n=391)	-3.84	41 : 26	0.09	1.48 (1.27)	7.14 (6.00)	0.06 (0.03)	2.53 (3.06)	18.57 (34.81)	81.11 (81.93)	0.70 (0.69)	0.00 (0.12)
prompt: counterfactual_invariance	explicit	8.67 [6.00, 11.33]	-3.33	51 : 36	0.13	1.00 (1.27)	3.78 (6.00)	0.04 (0.03)	2.02 (3.06)	17.43 (34.81)	78.37 (81.93)	0.68 (0.69)	0.12 (0.12)
prompt: zero_shot_cot	explicit	8.67 [6.00, 11.56]	-3.33	44 : 29	0.10	0.79 (1.27)	3.78 (6.00)	0.04 (0.03)	2.10 (3.06)	22.22 (34.81)	82.57 (81.93)	0.70 (0.69)	0.15 (0.12)
prompt: second_pass_verification	explicit	10.00 [6.67, 13.56]	-2.00	51 : 42	0.41	1.39 (1.27)	3.78 (6.00)	0.09 (0.03)	2.17 (3.06)	4.77 (34.81)	70.49 (81.93)	0.61 (0.69)	7.60 (0.12)
prompt: ignore_personal_info	explicit	10.89 [8.00, 14.00]	-1.11	26 : 21	0.56	1.38 (1.27)	4.67 (6.00)	0.04 (0.03)	2.59 (3.06)	36.89 (34.81)	81.73 (81.93)	0.70 (0.69)	0.00 (0.12) ♦
prompt: reasoning	explicit	12.22 [9.11, 15.33]	+0.22	29 : 30	1.00	1.46 (1.27)	5.11 (6.00)	0.06 (0.03)	3.01 (3.06)	29.98 (34.81)	81.63 (81.93)	0.69 (0.69)	0.00 (0.12) ♦
scrub: lexical*	explicit	0.22 [0.00, 0.67]	-11.78	53 : 0	2e-16	0.08 (1.27)	0.22 (6.00)	0.00 (0.03)	0.05 (3.06)	33.73 (34.81)	83.60 (81.93)	0.70 (0.69)	0.00 (0.12)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.